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|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>5ai</b>  | <p>Using potential divider rule,</p> $V_{BD} = \left( \frac{R_{BD}}{R_{BD} + 1.0} \right) \times 2.0$ $V_{BD} = \left( \frac{4.0}{4.0 + 1.0} \right) \times 2.0 = 1.6 \text{ V}$                                                                                                                                                                                                                | <p>C1</p> <p>A1</p> |
| <b>5aii</b> | <p>Since current in Cell Y is zero, <math>V_{BC} = \text{e.m.f. of cell Y}</math></p> $l = \left( \frac{1.5}{1.6} \right) \times 100 = 94 \text{ cm}$                                                                                                                                                                                                                                           | <p>C1</p> <p>A1</p> |
| <b>5b</b>   | <p>Replace <u>cell X</u> with another cell with <u>lower internal resistance</u><br/> OR<br/> Use a resistance wire of higher resistivity, keeping cross-sectional area and length unchanged<br/> OR<br/> Use a resistance wire of smaller cross-sectional area, keeping resistivity and length unchanged.<br/> OR<br/> Use resistance wire with a <u>higher resistance per unit length</u></p> | <p>M1</p>           |

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|           | By potential divider rule, <u>this increases the p.d. across BD</u> and allows a smaller value of $l$ (while keeping current in cell Z zero).                             | A1           |
| <b>5c</b> | (When current is zero,) $V_{BC} = \text{e.m.f. of cell Z}$<br>p.d. across internal resistance is zero hence terminal p.d. = e.m.f.<br><br>Distance $l$ remains unchanged. | M1<br><br>A1 |

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| <b>6a</b> | electric and magnetic fields normal to each other in the same region | M1 |
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